

★ WEEK 5 · PART 3 · PRACTICE

VaR and Pairs, Worked by Hand

Turn a Volatility Forecast Into a Risk Number, Then a Trade

GARCH Value-at-Risk, the Kupiec test, and conditional bands

Part 3's payoff, with the arithmetic shown. We turn a GARCH volatility forecast into a Value-at-Risk figure and its Expected Shortfall, back-test the number with the Kupiec test, and scale a pairs trade's thresholds with the spread's own conditional volatility.

3P

PART 3 · APPLICATIONS

From a Forecast to a Risk Desk

Case A: convert a GARCH volatility forecast into 95% and 99% Value-at-Risk and the Expected Shortfall beyond it.
Case B: validate the VaR with the Kupiec back-test, and scale a pairs trade's bands with the spread's conditional volatility. Then a Claude Code lab.

What Part 3 Told Us

Now we apply it - with numbers

1

VaR is volatility times a quantile

A GARCH volatility forecast becomes a loss you shouldn't exceed on a normal day - and it breathes as volatility clusters.

2

Expected Shortfall reads the tail

Beyond the VaR line, ES is the average loss when things do go wrong - always larger, and more honest about extremes.

3

Back-test, then scale the bands

The Kupiec test checks the breach rate, and the spread's own GARCH volatility sets pairs thresholds that widen and tighten.

Everything here is a calculator exercise: one VaR, one shortfall, one back-test, one band.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a GARCH volatility forecast.
- Compute the risk number, or validate it.
- Then act on it.

What you should watch

- How volatility scales into a dollar loss.
- How the Kupiec test judges the model.
- How conditional bands scale with risk.

Value-at-Risk From GARCH

A forecast becomes a risk number

A risk desk needs a single figure: how much could we lose tomorrow on a bad-but-normal day? GARCH gives the volatility forecast; multiplying by a quantile and the position size gives Value-at-Risk. We compute it at 95% and 99%, then the Expected Shortfall - the average loss once that line is breached

The setup

- A \$10M portfolio.
- GARCH forecasts tomorrow's vol at 1.5%.
- A roughly zero daily mean.

What we'll do by hand

- Scale volatility by the 95% and 99% quantiles.
- Multiply by the position.
- Then compute the shortfall.

Value-at-Risk

Volatility times a quantile

$$\text{VaR}_\alpha = z_\alpha \cdot \underbrace{\sigma_t}_{\text{GARCH forecast}} \cdot V \quad (\text{volatility} \times \text{quantile} \times \text{position})$$

The three factors

- The GARCH volatility forecast.
- The quantile of the loss distribution.
- The size of the position.

Why GARCH matters

- The volatility is conditional, not constant.
- So VaR rises after a shock, falls in calm.
- A static number can't do that.

The VaR at Two Levels

95% and 99%

$$95\% : 1.645 \times 0.015 \times \$10\text{M} = \$\mathbf{247\text{K}}$$

$$99\% : 2.326 \times 0.015 \times \$10\text{M} = \$\mathbf{349\text{K}}$$

At 95%, the quantile is 1.645, giving a one-day VaR of about \$247K. At 99%, the quantile rises to 2.326 and the VaR to \$349K. The higher the confidence, the larger the loss you're insuring against.

Expected Shortfall

The average loss in the tail

$$ES_{95\%} = \frac{\phi(z_\alpha)}{\alpha} \sigma_t V = \frac{0.1031}{0.05} \times 0.015 \times \$10M = \$309K (> VaR)$$

What it measures

- Not the threshold, but the average beyond it.
- The mean loss on the worst 5% of days.
- Here about \$309K.

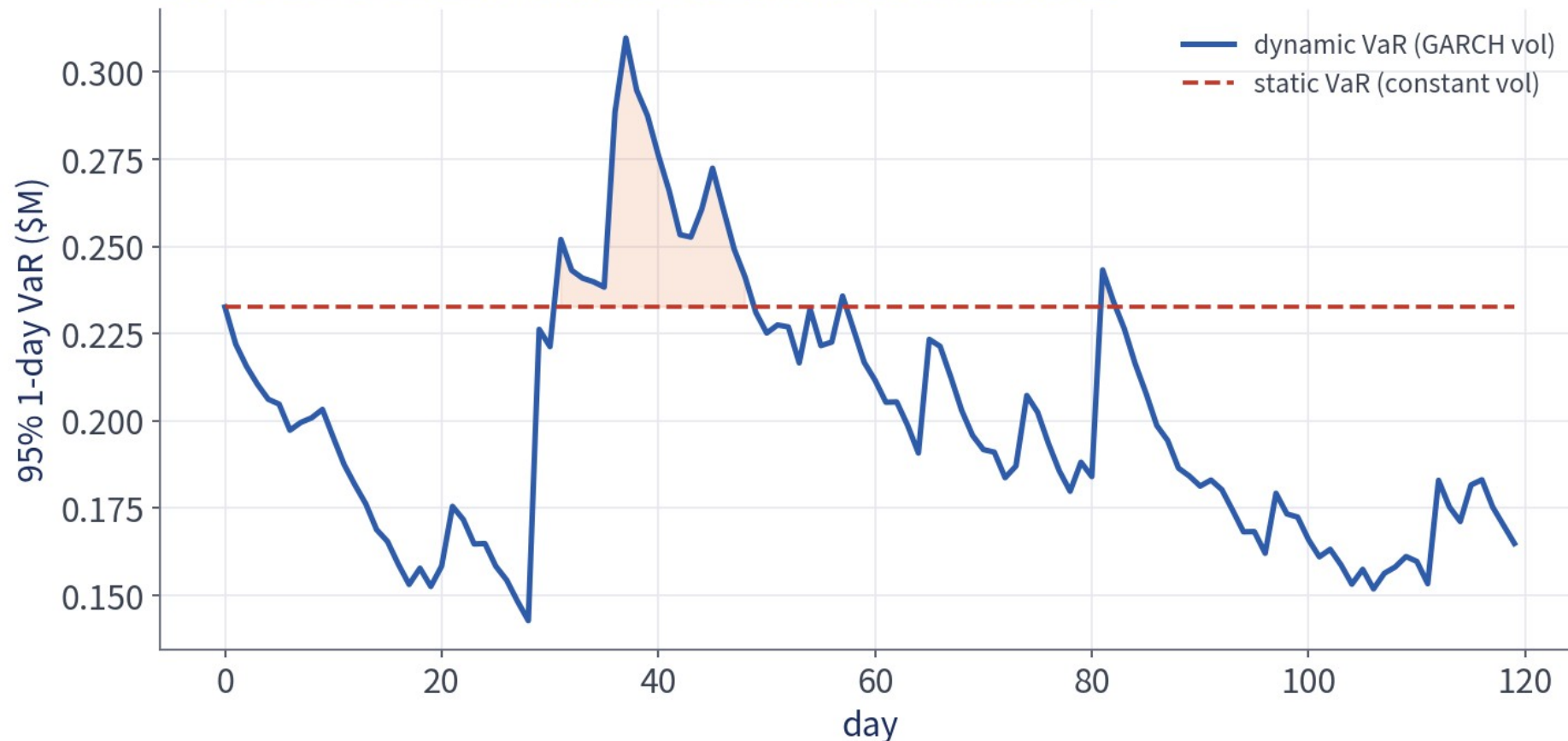
Why it's larger

- VaR is only the edge of the tail.
- ES averages the losses deeper inside it.
- So ES always exceeds VaR.

VaR That Breathes

The GARCH-based VaR (blue) rises sharply after each shock and relaxes in calm spells, tracking the conditional volatility. The static VaR (dashed red) is flat - and in the shaded periods it sits below the dynamic line, understating the true risk exactly when it matters most.

Case A: GARCH VaR breathes - it rises after shocks, when static VaR understates



A Living Risk Number

Why static VaR is dangerous

The static trap

- One VaR for all seasons.
- It understates risk after a shock.
- Breaches cluster right when they hurt.

The GARCH fix

- VaR scales up as volatility clusters.
- It relaxes when markets calm.
- ES then sizes the tail beyond it.

Validate, Then Trade

Back-test the VaR, scale the bands

A risk number is only as good as its track record. The Kupiec test checks whether the VaR was breached at the right rate - too often means the model understates risk. Then we bring GARCH to the pairs trade of Weeks 6 and 7: the spread has its own conditional volatility, so its trading bands should widen and tighten with it

The validation

- Count the VaR breaches over 250 days.
- Compare to the expected 12.5.
- The Kupiec test judges the gap.

What we'll do by hand

- Run Kupiec on a good and a bad model.
- Then build conditional bands on the spread.
- Compare to static thresholds.

The Kupiec Test

Is the breach rate correct?

$$LR = 2 \left[x \ln \frac{x/N}{p} + (N - x) \ln \frac{1-x/N}{1-p} \right] \sim \chi_1^2 \quad (\text{is the breach rate correct?})$$

The idea

- A 95% VaR should be breached about 5% of days.
- Count the actual breaches.
- Test whether the rate is statistically right.

The verdict

- The statistic follows a chi-square.
- Below 3.84, the model passes.
- Above, too many breaches - reject it.

Two Models, Two Verdicts

The test catches the bad one

breaches (of 250)	rate	LR statistic	verdict
14 (expected 12.5)	5.6%	0.18	PASS
25	10.0%	10.33	REJECT

25 model with 14 breaches - close to the expected 10.33 - gives a Kupiec statistic of 0.18, well under 3.84: it passes. But 25 breaches, a 10% rate, pushes the statistic to 10.33 and is firmly rejected - that VaR understates risk.

Conditional Bands on the Spread

Thresholds that scale with risk

spread bands = $\bar{s} \pm k \sigma_{s,t}$ (GARCH) : wide when volatile, tight when calm

The idea

- The spread's volatility clusters too.
- Fixed z-score bands ignore that.
- GARCH bands scale with the spread's risk.

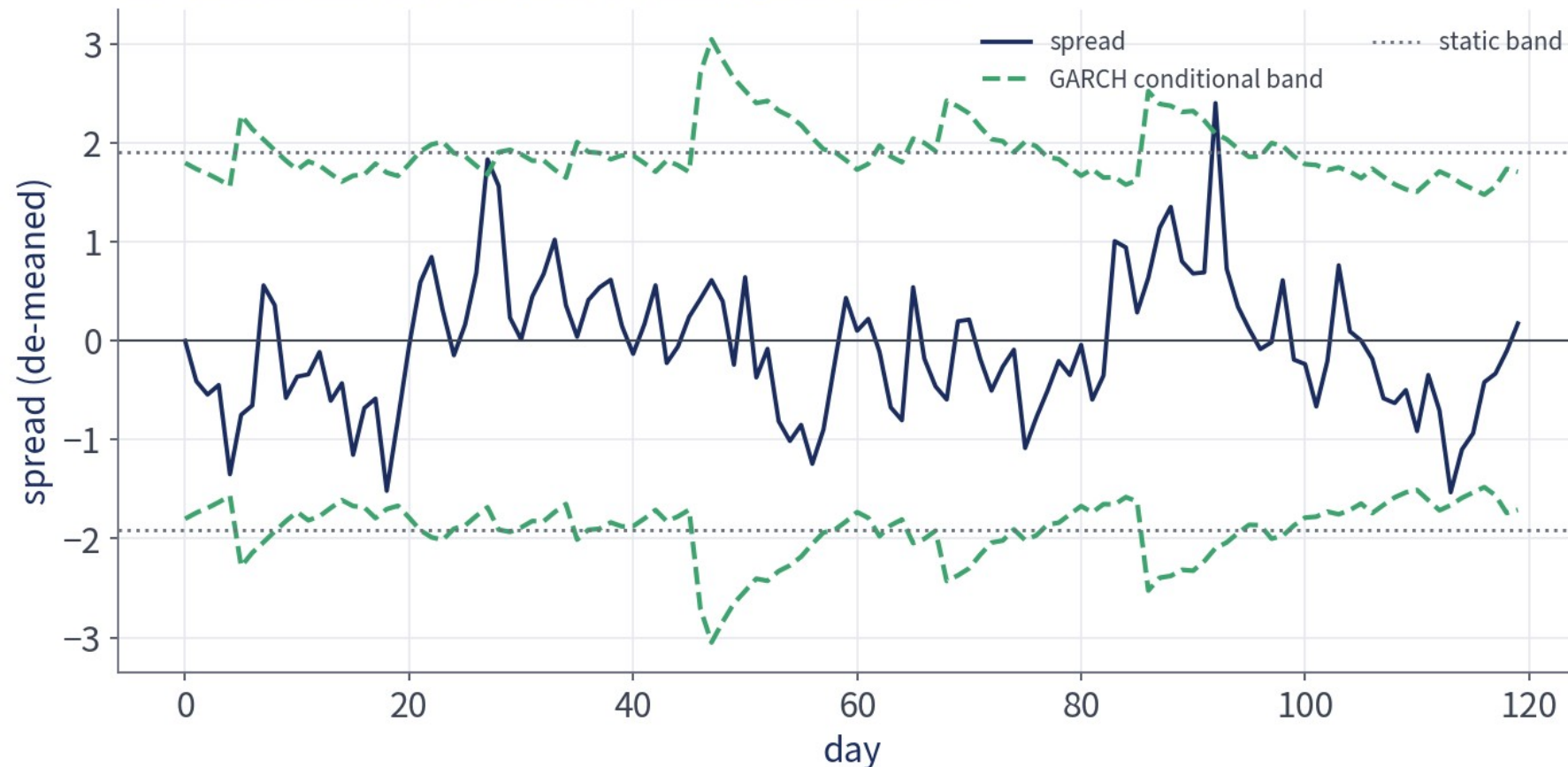
The payoff

- Wider bands when the spread is turbulent.
- Tighter bands when it's calm.
- Fewer false signals, better-timed trades.

Bands That Move With Risk

The GARCH conditional bands (green) breathe with the spread's volatility - flaring out during turbulent stretches and drawing in when it's quiet. Static bands (dotted) stay fixed, triggering too often in calm and too rarely in storms. Scaling with risk times the pairs trade better.

Case B: bands widen when the spread is volatile, tighten when calm



Trust, but Verify

Validation and dynamic thresholds

Why back-test

- A VaR that breaches too often is false comfort.
- Kupiec turns the breach count into a verdict.
- Regulators and desks both demand it.

Why conditional bands

- The spread's risk isn't constant.
- Static bands misfire across regimes.
- GARCH bands adapt, trade by trade.

The Week's Arc

From uncovering a risk

Week 5 built a full pipeline. ARCH saw that variance is conditional; GARCH modeled it parsimoniously and made it forecastable; Value-at-Risk turned that forecast into a dollar figure; the Kupiec test validated it; and the same conditional volatility scaled a pairs trade's bands. One idea - risk that clusters and mean-reverts - runs

The risk number (Case A)

- 95% VaR \$247K, 99% \$349K.
- Expected Shortfall \$309K.
- It breathes with volatility.

The validation (Case B)

- Kupiec passes at 14 breaches, fails at 25.
- Conditional bands scale with the spread.
- Forecast, validated, and traded.

Reproduce the VaR and the Back-Test

Direct Claude Code to check your arithmetic

GOAL Compute VaR and ES from a GARCH forecast, then back-test with Kupiec.

[INPUT] Prompt · Claude Code

```
Write and run part3_practice.py that:  
CASE A - VaR and ES:  
  * with V=$10M and a 1.5% GARCH vol forecast,  
    print the 95% and 99% VaR;  
  * print the 95% Expected Shortfall.  
CASE B - back-test:  
  * for N=250 and breaches x=14 and x=25, print the  
    Kupiec LR statistic and compare to 3.84;  
  * build GARCH conditional bands on a spread.  
Then plot the dynamic VaR and the conditional bands.
```

Check against the slides

- 95% VaR \$247K, 99% \$349K.
- ES \$309K > VaR.
- Kupiec 0.18 pass, 10.33 reject.
- bands scale with vol.

Deliverable: part3_practice.py whose printed numbers match the VaR, the ES, and the Kupiec verdicts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How does a GARCH forecast become VaR?

Volatility times a quantile times the position

Q2 Why is Expected Shortfall larger than VaR?

It averages the losses beyond the VaR line

Q3 What does the Kupiec test check?

Whether the VaR breach rate is statistically correct

All three come from the VaR formula, the Expected Shortfall, and the Kupiec back-test.

Three Pitfalls to Avoid

Where VaR and pairs trip people

Common mistakes

- "A static VaR is good enough."
- "VaR tells you the worst-case loss."
- "Fixed z-score bands suit every regime."

The reality

- Static VaR understates risk after shocks.
- VaR is a threshold; ES sizes the tail beyond it.
- Scale pairs bands with the spread's volatility.

Two Calculations to Remember

Part 3 applications in one screen

1

A forecast becomes a risk number

A 1.5% GARCH volatility forecast on a \$10M book gave a 95% VaR of \$247K and a 99% VaR of \$349K, with an Expected Shortfall of \$309K in the tail beyond. Because the volatility is conditional, the VaR breathes - rising after shocks, relaxing in calm.

2

Validate the number, then trade on it

The Kupiec test passed a model with 14 breaches (statistic 0.18) and rejected one with 25 (statistic 10.33). The same conditional volatility scaled a pairs trade's bands - wide when the spread is turbulent, tight when it's calm.

3

One idea runs through the week

Risk that clusters and mean-reverts drove ARCH, GARCH, Value-at-Risk, its back-test, and the pairs bands alike - a single volatility model turned into a working risk desk.

Week 5 complete — from ARCH and GARCH to a validated Value-at-Risk and a volatility-timed pairs trade.